<table>
<tr><td>Project Title</td><td colspan="2">: Reasoning-Guided Multimodal Captioning of Satellite Images Using Contrastive Language–Image Pre-training</td></tr>
<tr><td rowspan="3">Industry Partner / Association</td><td>Organization / Company Name</td><td>:</td></tr>
<tr><td>Industry Domain:
(e.g., Healthcare, FinTech, Manufacturing, AI, EdTech, etc.)</td><td>: Remote Sensing</td></tr>
<tr><td>Industry Mentor Details
(Name, Phone and Email – Optional)</td><td>: NA</td></tr>
<tr><td>Problem Statement
(Brief description of the real-world problem provided by the industry. Clearly state the challenge, context, and what needs to be solved. Keep it concise and understandable.)</td><td colspan="2">Remote Sensing Image Captioning (RSIC) plays a crucial role in automatically interpreting satellite imagery for applications such as urban planning, environmental monitoring, and disaster management. Existing approaches, including Contrastive Language–Image Pre-training (CLIP) guided feature fusion and attention-based frameworks, have improved visual-semantic alignment but often lack explainability and explicit reasoning capabilities. This research proposes a Chain-of-Thought CLIP Framework that integrates CLIP-based multimodal representations with reasoning-guided caption generation to model intermediate scene-understanding steps before producing descriptions. The proposed framework aims to enhance semantic understanding, reduce caption hallucinations, and improve interpretability. Expected outcomes include more accurate, context-aware, explainable, and trustworthy satellite image captions.</td></tr>
<tr><td>Project Objectives / Expected Outcome
(What students are expected to achieve by the end of the project)</td><td colspan="2">Project Objectives
Develop a CLIP-based multimodal framework for remote sensing image captioning.
Integrate Chain-of-Thought reasoning for explainable scene interpretation.
Enhance visual-semantic representations through reasoning-guided feature refinement.
Minimize semantic hallucinations using reasoning-consistency constraints.
Evaluate the proposed framework against state-of-the-art captioning models on benchmark datasets
Expected Outcome
Development of an explainable CLIP-based remote sensing image captioning framework that effectively aligns visual and textual information.</td></tr>
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- Generation of interpretable Chain-of-Thought reasoning sequences that reveal the decision-making process behind caption generation.
- Enhanced semantic understanding of satellite scenes through reasoning-guided multimodal feature learning.
- Reduction of semantic hallucinations and increased caption reliability, leading to more trustworthy remote sensing image descriptions.

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<tr><td>Proposed Solution Scope
(High-level scope of work — what students will design, develop, analyze, or implement)</td><td colspan="3">The proposed Chain-of-Thought Guided CLIP Framework for Explainable Remote Sensing Image Captioning enables accurate and interpretable analysis of satellite imagery, benefiting sectors such as urban planning, agriculture, environmental monitoring, disaster management, defense, and geospatial intelligence. By generating explainable captions, it supports transparent decision-making and efficient utilization of large-scale remote sensing data. For society, the framework contributes to improved disaster response, resource management, and sustainable development. Students gain practical experience in Artificial Intelligence, Computer Vision, Natural Language Processing, Multimodal Learning, and Explainable AI, enhancing their technical expertise, research capabilities, and readiness for careers in industry and advanced research.</td></tr>
<tr><td colspan="2">Problem Complexity / Difficulty Level
(Beginner / Intermediate / Advanced)</td><td colspan="2">: Intermediate</td></tr>
<tr><td rowspan="2">Technology Stack
(Indicative tools and technologies to be used)</td><td>Programming Languages</td><td colspan="2">: Python</td></tr>
<tr><td>Frameworks / Libraries</td><td colspan="2">

- PyTorch – Main deep learning framework for implementing CLIP, Chain-of-Thought modules, and captioning models.
- Hugging Face Transformers – For pretrained vision-language models and language models.
- OpenCLIP – Open-source implementation of CLIP for vision-language representation learning.
- Sentence Transformers – For text embedding and semantic similarity analysis.

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	Databases / Platforms	RS Captioning Dataset	
	Tools / APIs (if any)		
COE (Center of Excellence) Mapping	Is this project mapped to a COE? (Yes / No)		: No
	If Yes, COE Name		:
Faculty Coordinator *(who sourced Project)*	Faculty Name		: Dr. A.EZIL SAM LENI
Student Engagement Details ***(Optional)***	Target Year / Program		: III / IV Year – CSE/ AIML
	Team Size		: 3 – 4 Students
	Project Duration		: 6 – 12 Months

Industry Value / Impact *(How this project benefits the industry partner and enhances student industry readiness)*	Students are expected to design and develop an explainable remote sensing image captioning framework that integrates Contrastive Language–Image Pre-training (CLIP) with Chain-of-Thought (CoT) reasoning for generating accurate and interpretable captions from satellite images. The work involves implementing multimodal feature extraction, reasoning-guided attention mechanisms, and caption generation modules capable of producing intermediate reasoning steps alongside final descriptions. Students will analyze the effectiveness of the proposed framework in improving semantic understanding, reducing caption hallucinations, and enhancing explainability. They will also evaluate and compare the model's performance against existing state-of-the-art remote sensing image captioning approaches using benchmark datasets and standard captioning metrics
Expected Deliverables *(Functional prototype / model, Technical report, Presentation / demo)*	1. Research Output 2. Demo / Presentation
IP / NDA Requirement: *(If Yes, details to be shared separately)*	: Optional
Project Alignment - Mapped Course(s) if any	Machine Learning, Generative AI